

DAY 3 TASK 3

MASTERING ENCAPSULATION AND ACCESS CONTROL

Learn how to **protect data** using encapsulation, how to define **private variables**, and how to use **getter and setter methods**.

✓ Assigned Task 3

◆ Theoretical Concepts You Must Learn Today:

1. What is **Encapsulation** in OOP?
 2. What are **Access Modifiers** (public, private, protected) in Python?
 3. Why and when should we **hide data** in real-world projects?
 4. What are **getter and setter methods**? Why are they important?
-

◆ Practical Coding Task:

➤ Scenario:

You're creating a **BankAccount** class where:

- The balance should not be directly accessible or changeable.
- Only controlled functions should be able to **set** or **get** the balance.

❖ Task Requirements:

1. Create a class called BankAccount.
2. Use `__init__()` to take:
 - `account_holder`
 - `initial_balance` (set this as a **private** variable using `__balance`)
3. Add the following methods:
 - `deposit(amount)` —> only adds if amount is positive
 - `withdraw(amount)` —> checks if balance is enough before deducting
 - `get_balance()` —> returns current balance (getter)
 - `set_balance(amount)` —> sets balance only if amount is valid (setter with validation)
4. Create an object with sample data.

5. Try to **access __balance directly** (and see it fails).

6. Use the methods to:

- Deposit money
- Withdraw money
- Print balance using getter

📌Your Output Example.... :

Depositing 500...

Withdrawing 200...

Current balance: 300

➔ Bonus Twist (Optional):

- Try to create 2–3 objects with different account holders.
 - Block deposit of negative amounts using a condition.
 - Add method `account_summary()` that prints:
 - Account Holder: Ali
 - Current Balance: 300
-

❖ Bonus Tips:

- Use `self.__balance` for private variable.
 - Use meaningful method names (helps you in your viva).
 - Don't forget: Encapsulation is about **control and protection**.
-

🕒 **Deadline:**

Submit by: *Tomorrow before 11:59 PM*

Day: 3 of 8

You're building strong foundations..... consistency is your secret weapon ⌚.

 **Final:**

After completing, reply with:

“ Task 3 Completed “

And then I’ll guide you to **Day 4’s** powerpacked OOP task.

Keep smashing it Future Agentic AI Developer in the making..... 
